



Republic of Iraq
Ministry of Higher Education and
Scientific Research
University of Al Qadisiyah
College of Computer Science &
Information Technology
Information Systems Department



A Template-Based Website Builder

A project Submitted To Council of the College of Computer Science
& Information Technology, Information Systems Department ,
University of Al Qadisiyah in Partial Fulfilment of the Requirements
for the BSc Degree in Information Systems

By

Mohammad Ali Flayyih

Supervised By

Dr. Alaa Abid Muslim Abid Ali

April 2026



(يَرْفَعُ اللَّهُ الَّذِينَ آمَنُوا مِنْكُمْ وَالَّذِينَ أُوتُوا الْعِلْمَ دَرَجَاتٍ)

[المجادلة: 11]



Dedication

To my beloved family. . .

My Mother

My Father

My Sisters

To my lovely friends

And everyone who cares . . .

Acknowledgments

In preparing this report, I was in contact with many people, researchers, academicians, and practitioners. They have contributed towards my understanding and thought. In particular, I wish to express my sincere appreciation to my main project supervisor, Dr. Alaa Abid Muslim Abid Ali, for encouragement, guidance critics and friendship. I am also very thankful to all my professors for their guidance, advices and motivation. Without their continued support and interest, this project would not have been same as presented here

Librarians at university of Al Qadisiyah. I am also indebted to Information Systems Department faculty; my university also deserve special thanks for their assistance in supplying the relevant literatures.

Supervisor Certification

I certify that the project entitled “A Template-Based Website Builder” was prepared under my supervision at the Information Systems Department / College of Computer Science & Information Technology /University of Al Qadisiyah, as partial fulfillment of the requirements needed for award of the degree of Bachelor in Information Systems.

Signature:

Name: Dr. Alaa Abid Muslim Abid Ali

Date: / / 2026

Email: alaa.abidmuslam@qu.edu.iq

Examination Committee Certification

I hereby certify that this project entitled "A Template-Based Website Builder" Was prepared under my supervision at the Information Systems Department / College of Computer Science & Information Technology /University of Al Qadisiyah, as a partial fulfillment of the requirements for the Bachelor in Information Systems.

(Chairman)

Name:

Date: / / 2026

Signature:

(Member)

Name:

Date: / / 2026

Signature:

Table of Contents

Contents	Page No.
List Of Tables	ix
List Of Figures	x
List Of Abbreviations	xi
Abstract	xii
Chapter One: Introduction	
1.1 Introduction	1
1.2 The Problem Statement	2
1.3 Research Objectives	3
1.4 Project Organization	4
Chapter Tow: Theoretical background	
2.1 Introduction	5
2.2 Languages	5
2.3 Frameworks and Libraries	6
2.4 Development Tools	9
2.5 Database Design	11

Chapter Three: Design and Implementation	
3.1 Design and Implementation	17
3.2 General Structure	17
3.3 User Interfaces and Pages	20
3.4 State Management and Algorithmic Logic	31
Chapter Four: Conclusions and Future Works	
4.1 Conclusions	34
4.2 Future Works	35
References	
References	37

LIST OF TABLES

Table 1: User Collection.....	12
Table 2: Project Collection.....	14
Table 3: Template Collection.....	16

LIST OF FIGURES

Figure 1: Illustrated for system architecture.....	17
Figure 2: User Authentication.....	20
Figure 3: User Registration Interface.....	21
Figure 4: Full Editor Workspace Structure (Sidebar, Canvas,Properties).....	24
Figure 5: Preview Mode simulating final website rendering.....	26
Figure 6: The internal Template Gallery selection screen.....	28
Figure 7: About Page.....	30

LIST OF ABBREVIATIONS

Abbreviation	Full Term
AI	Artificial Intelligenc
API	Application Programming Interface
BSON	Binary JavaScript Object Notation
CSS	Cascading Style Sheets
CORS	Cross-Origin Resource Sharing
DB	Database
DOM	Document Object Model
FPS	Frames Per Second
HTML	HyperText Markup Language
IDE	Integrated Development Environment
I/O	Input/Output
JSON	JavaScript Object Notation
JS	JavaScript
JWT	JSON Web Token
MERN	MongoDB, Express.js, React, Node.js
NoSQL	Not Only SQL (Non-relational database)
REST	Representational State Transfer
SEO	Search Engine Optimization
SPA	Single Page Application
SQL	Structured Query Language
TWB	Template Website Builder
UI	User Interface
UX	User Experience
VCS	Version Control System
VS Code	Visual Studio Code
WYSIWYG	What You See Is What You Get

Abstract

Establishing a robust and visually compelling web presence has become a fundamental necessity for businesses, creators, and individuals in the contemporary digital era. Despite this, web development remains a polarized field, often divided between professional coding environments that require deep technical expertise and traditional drag-and-drop builders that sacrifice creative freedom and performance for ease of use. This research project introduces the TWB — Template Website Builder, a pioneering platform designed to bridge this gap by offering a high-performance, WYSIWYG (What You See Is What You Get) workspace that prioritizes both user autonomy and technical excellence.

The centerpiece of the TWB project is its innovative Hybrid Positioning Engine. Unlike standard builders that confine users to rigid grid systems, TWB enables a seamless transition between freeform absolute positioning allowing for precise, pixel-perfect design and structured responsive flows that ensure cross-device compatibility. This approach provides the intuitive feel of professional graphic design tools while maintaining the structural integrity and performance of a modern website. Furthermore, the platform addresses the pervasive issue of "code bloat" by generating clean, optimized, and production-ready code in real-time.

Leveraging a robust technical stack comprising React.js, Node.js, and MongoDB, TWB implements advanced features such as an infinite canvas workspace and real-time interactive state previewing. The research concludes that by integrating these professional-grade design capabilities into a no-code environment, TWB successfully democratizes high-end web development. It provides an efficient solution for users seeking to create sophisticated, responsive, and performance-driven websites without the steep learning curve traditionally associated with professional web design tools.

Keywords: Web Development, No-Code, Drag-and-Drop, WYSIWYG, Hybrid Positioning Engine, React.js, Responsive Design, Website Builder.